Practical: Implement Job Sequencing using Greedy Method.

What is it?

Suppose you have 4 jobs.

Every job has:

- **Deadline = Last time to complete the job.**
- **Profit = Money earned after completing the job.**

Our aim is to earn maximum profit.

What is Greedy Method?

Greedy means choosing the best option available at the current step.

Steps:

1. **Sort jobs according to profit (highest first).**
2. **Select the job with highest profit.**
3. **Put it in the latest available slot before its deadline.**
4. **Repeat for remaining jobs.**
5. **Calculate total profit.**

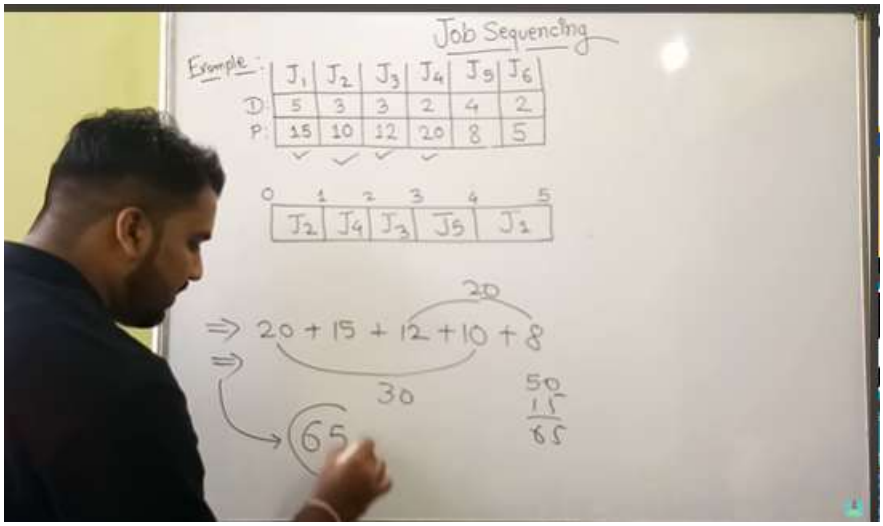
For the above example:

- **J3 → Slot 2 → Profit 200**
- **J1 → Slot 1 → Profit 100**
- **J2 → Cannot fit.**
- **J4 → Cannot fit.**

Maximum Profit = 300

Complexity: For the simple sorting + slot-search implementation:

- **Time: O(n log n + nD), where D is the maximum deadline.**
- **Space: O(D) for the slots.**



BC-4: Survey Report on Types of Blockchains and Their Real-Time Use Cases

Blockchain Technology Practical | Simple Baby English

1. Aim

To study different types of blockchains and their real-time uses.

2. Introduction

Blockchain is a digital technology used to store data.

- It stores data in blocks.
- These blocks are connected to each other.
- Data is stored safely.
- Data cannot be easily changed or deleted.
- It does not need one central person to control everything.

Blockchain is used in banking, healthcare, education, shopping and many other fields.

3. Types of Blockchain

There are mainly four types of blockchain.

1. Public Blockchain

- Anyone can join this blockchain.
- Anyone can see the transactions.
- It is open to everyone.
- It is very transparent.

Examples: Bitcoin and Ethereum.

Real-time uses:

- Sending and receiving cryptocurrency.
- Online payments.
- Smart contracts.
- Digital transactions.

2. Private Blockchain

- It is controlled by one organization.
- Only allowed people can join.
- Data is private and secure.
- It is mostly used by companies.

Examples: Hyperledger Fabric (can be used to build private networks).

Real-time uses:

- Company records.
- Employee information.
- Banking records.
- Business transactions.

3. Consortium Blockchain

- It is controlled by many organizations.
- Only selected organizations can join.
- All members share the data.
- It is useful for business work.

Examples: R3 Corda and Quorum.

Real-time uses:

- Banking systems.
- Supply chain management.
- Insurance companies.
- Sharing data between hospitals.

4. Hybrid Blockchain

- It is a combination of public and private blockchain.
- Some data is public.
- Some data is private.
- It gives both privacy and transparency.

Real-time uses:

- Healthcare records.
- Government records.
- Real estate.
- Supply chain management.

4. Comparison of Blockchain Types

Type	Who can join?	Example
Public	Anyone	Bitcoin
Private	Selected people	Hyperledger Fabric
Consortium	Selected organizations	R3 Corda
Hybrid	Public + selected people	Hybrid blockchain systems

5. Real-Time Applications of Blockchain

1. Banking: Used for fast and secure money transfers.
2. Healthcare: Used to store patient records safely.
3. Education: Used to store certificates and marksheets.
4. Supply Chain: Used to track products from factory to customer.
5. Voting: Can be used for secure digital voting.
6. Real Estate: Used to store property records.
7. Cryptocurrency: Used for digital currencies like Bitcoin.

6. Advantages of Blockchain

- Data is secure.
- Transactions are transparent.

- Data is difficult to change.
- No central control is needed in decentralized networks.
- It reduces fraud.
- It builds trust between users.

7. Disadvantages of Blockchain

- Some blockchains are slow.
- Setup can be costly.
- It needs technical knowledge.
- Some blockchains use more electricity.
- Transactions may take time.

8. Conclusion

Blockchain is a useful technology for storing data safely. There are four main types: Public, Private, Consortium and Hybrid Blockchain. Each type has different uses. Blockchain is used in banking, healthcare, education, voting and many other fields. It helps to make transactions safe and transparent.

9. Viva Questions

Q1. What is Blockchain?

Blockchain is a digital system used to store data safely.

Q2. How many types of blockchain are there?

There are four main types.

Q3. Name the four types of blockchain.

Public, Private, Consortium and Hybrid.

Q4. What is Public Blockchain?

It is open to everyone.

Q5. What is Private Blockchain?

It is controlled by one organization.

Q6. What is Bitcoin?

Bitcoin is a digital currency that uses blockchain.

Q7. What is Ethereum?

Ethereum is a blockchain platform that supports smart contracts.

Q8. What is the main advantage of blockchain?

It provides security and transparency.

Q9. Where is blockchain used?

It is used in banking, healthcare, education and supply chains.

Q10. What is a Hybrid Blockchain?

It is a combination of public and private blockchain.

Practical ready

Tip: Write this report in your practical journal with the diagrams. The language is kept simple so you can easily understand and explain it during your viva. ❤️‍🩹

Q3) HUFFMAN ENCODING =LOSS less data compression algorithm

Practical: Implement Huffman Encoding using Greedy Strategy.

What is Huffman Encoding?

Huffman Encoding is used to compress data.

It converts characters into binary codes (0 and 1).

Example:

Suppose we have:

Character	Frequency
A	5
B	9
C	12
D	13

Frequency means how many times a character appears.

How does it work?

1. Find the two characters with the lowest frequency.
2. Combine them.

3. Add their frequencies.
4. Repeat until one tree is created.
5. Give 0 to the left branch and 1 to the right branch.
6. Get the binary code for each character.

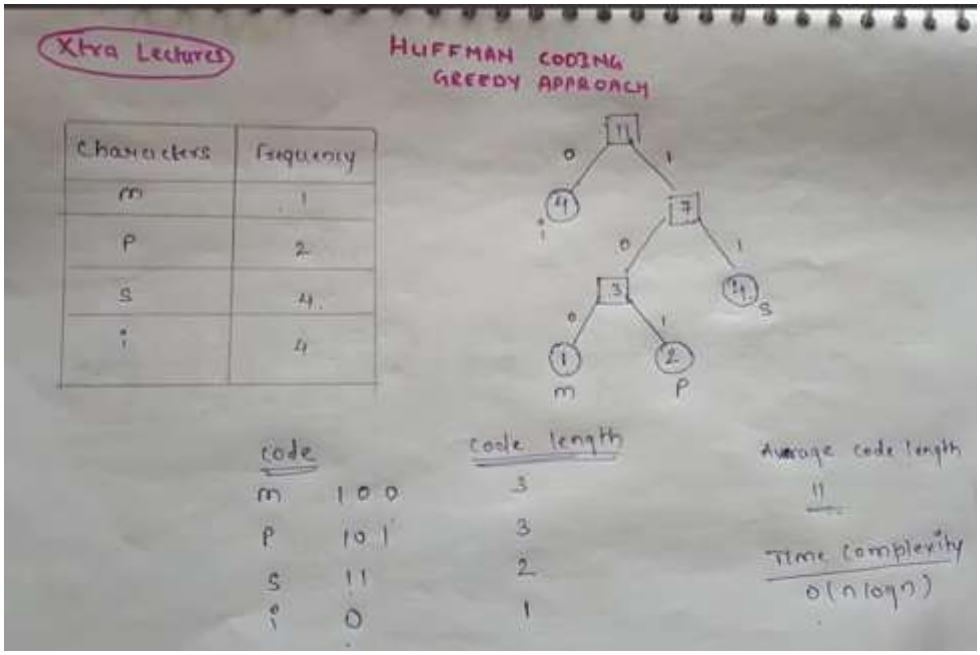
Important: Characters used more often generally get shorter codes.

Example: A character may get 0, another may get 10, another may get 110.

The exact codes depend on the tree and branch choices.

Complexity:

- Time: O(n log n)
- Space: O(n)



Q1. What is Huffman Encoding?

It is a data compression technique.

Q2. Why do we use Huffman Encoding?

To reduce storage space.

Q3. Which technique does Huffman use?

Greedy Method.

Q4. What is frequency?

How many times a character appears.

Q5. What is a binary tree?

A tree where each node has at most two children.

Q6. Which character gets the shortest code?

The character with the highest frequency generally gets a shorter code.

Q7. What are 0 and 1 used for?

They represent the left and right branches of the tree.

4. Fractional Knapsack

practical mein ek simple concept hai: Bag mein items daalke maximum profit earn karna.

1. What is Fractional Knapsack?

Knapsack means Bag.

Suppose:

- Tere paas ek bag hai.
- Bag ki capacity limited hai.
- Tere paas different items hain.
- Har item ka weight aur profit hai.

Tujhe bag mein items daalne hain so that maximum profit mile.

Fractional means: Tu kisi item ka half, quarter ya 25% bhi le sakti hai.

2. Understand the example

Suppose bag capacity = 50 kg.

We have 3 items:

Item	Weight	Profit
A	10 kg	₹60
B	20 kg	₹100

C	30 kg	₹120

Our aim is to get maximum profit without exceeding 50 kg.

3. Step 1: Calculate Profit / Weight Ratio

First, divide profit by weight.

\[Ratio = \frac{Profit}{Weight} \]

Item	Weight	Profit	Ratio
A	10	60	6
B	20	100	5
C	30	120	4

Why do we calculate ratio?

Ratio tells us how much profit we get for 1 kg of weight.

For example:

- A gives ₹6 profit per kg.
- B gives ₹5 profit per kg.
- C gives ₹4 profit per kg.

So A is the best item.

4. Step 2: Sort according to Ratio

Arrange items from highest ratio to lowest ratio.

Priority	Item	Ratio
1	A	6
2	B	5
3	C	4

So our order is:

A → B → C

5. Step 3: Put items into the bag

Bag capacity = 50 kg.



7. What is Greedy Method?

This is very important for viva.

Greedy means **choosing the best option at every step.**

In our example:

1. First choose A because ratio is 6.
2. Then choose B because ratio is 5.
3. Then choose C because ratio is 4.
4. Take a fraction of C to fill the bag.

This is called Greedy Method.

8. Algorithm (Write in Practical Journal)

1. Start.
2. Read the number of items.
3. Read weight and profit of each item.
4. Calculate profit/weight ratio.
5. Sort items in descending order of ratio.
6. Select items with the highest ratio first.
7. If the complete item fits, take it.
8. Otherwise, take only the required fraction.
9. Calculate total profit.
10. Stop.

9. Time and Space Complexity

Complexity	Answer
Time Complexity	O(n log n)
Space Complexity	O(n) if using an extra sorted array

Here, n = number of items.

9. Time and Space Complexity

Complexity	Answer
Time Complexity	O(n log n)
Space Complexity	O(n) if using an extra sorted array

Here, n = number of items.

10. Difference: Fractional vs 0/1 Knapsack

Fractional Knapsack	0/1 Knapsack
Can take half item	Cannot take half item
Uses Greedy	Uses Dynamic Programming
Sort by ratio	Uses DP table
O(n log n)	O(n × W)

11. Important Viva Questions

Q1. What is Fractional Knapsack?

It is a problem where we select full or partial items to get maximum profit.

Q2. Which method is used?

Greedy Method.

Q3. What is the formula for ratio?

Profit / Weight.

Q4. Why do we sort items?

To select the highest profit per unit weight first.

Q5. Can we take half an item?

Yes.

ML-1: Bank Customer Churn Prediction using Neural Network

1. First understand the practical

Imagine you have a bank.

The bank has 10,000 customers.

Some customers will continue using the bank, and some customers will leave the bank.

The bank wants to know:

Which customer will leave the bank in the next 6 months?

So, we build a Machine Learning model that predicts whether a customer will leave or stay.

This is called Bank Customer Churn Prediction.

What is Customer Churn?

Churn means leaving.

Example:

- You have an account in SBI.
- You close your SBI account.
- You move to HDFC Bank.

This is called Customer Churn.

Examiner asks: What is customer churn?

Say:

"Customer churn means when a customer stops using a company's service."

2. What is the output?

Our model gives two answers:

Output	Meaning
0	Customer will stay
1	Customer will leave

Example:

Our model predicts:

Customer A

0 – Stay

Customer B

1 – Leave

3. What is Neural Network?

This is the MOST IMPORTANT concept in ML-1.

Neural Network is a Machine Learning algorithm.

It works like our human brain.

Our brain takes information, understands it and gives an answer.

Similarly, a Neural Network:

1. Takes input.
2. Learns patterns.
3. Gives prediction.

It has 3 main layers.

ayer 1: Input Layer

It takes information from the dataset.

Example:

- Age
- Gender
- Credit Score
- Balance
- Tenure

Layer 2: Hidden Layer

This layer learns patterns from the data.

For example:

It can learn that customers with low credit scores or certain account behaviours may have a higher chance of leaving.

Layer 3: Output Layer

It gives the final answer.

- 0 = Stay
- 1 = Leave

Examiner asks: What is Neural Network?

Say:

"Neural Network is a Machine Learning algorithm inspired by the human brain. It takes input, learns patterns and gives output."

4. Understand our dataset

Our dataset is called Churn_Modelling.csv.

It contains 10,000 customer records.

Some important columns:

Column	Meaning
CustomerId	Customer's ID
CreditScore	Customer's credit score
Geography	Customer's country
Gender	Male or Female
Age	Customer's age
Tenure	Years with bank
Balance	Money in bank account
NumOfProducts	Number of bank products
IsActiveMember	Active customer or not
EstimatedSalary	Customer's estimated salary
Exited	Customer left or not

Remember: Exited is our target column.

- Exited = 0 means customer stayed.
- Exited = 1 means customer left.

ML-2: Email Spam Classification using KNN and SVM

KNN full form is K-Nearest Neighbors.

It is a supervised Machine Learning algorithm.

It classifies new data by checking its nearest neighbors.

Understand with an example

Suppose a new email comes.

Our model checks its 5 nearest similar emails.

Among those 5 emails:

- 3 are Spam.
- 2 are Not Spam.

What will our model predict?

Answer: Spam.

Because Spam has more votes.

This is called Majority Voting.

How KNN works?

1. Select K value.
2. Calculate distance between data points.
3. Find nearest K neighbors.
4. Check majority class.
5. Give final prediction.

Important terms

What is K?

K means number of nearest neighbors.

Example: K = 5 means we check 5 nearest neighbors.

What is Euclidean Distance?

It calculates the straight-line distance between two data points.

What is Majority Voting?

The class having the highest number of votes becomes the final prediction.

What is SVM?

SVM full form is Support Vector Machine.

It is also a supervised Machine Learning algorithm.

It separates two classes using a boundary.

Understand with an example

Imagine we have two groups:

- Spam emails.
- Normal emails.

SVM tries to create a line that separates these two groups.

This line is called a Hyperplane.

Important terms

1. Hyperplane

A line or boundary that separates two classes.

2. Margin

Distance between the separating boundary and the nearest data points from both classes.

3. Support Vectors

The nearest data points to the separating boundary.

4. Kernel

Kernel helps SVM separate data when a simple straight line is not enough.

How SVM works?

1. Takes input data.
2. Finds the best separating boundary.
3. Separates Spam and Not Spam.
4. Predicts the class of new emails.

Examiner asks: What is SVM?

Say:

"SVM is a supervised Machine Learning algorithm that separates different classes using the best possible decision boundary."

6. KNN vs SVM

This is very important because your practical asks you to analyze their performance.

KNN	SVM
K-Nearest Neighbors	Support Vector Machine
Uses nearest neighbors	Uses separating boundary
Uses majority voting	Uses hyperplane
K value is important	Kernel is important
Can be slow during prediction on large datasets	Can work well with high-dimensional data

Examiner: Which is better, KNN or SVM?

Say:

"It depends on the dataset. We compare both algorithms using accuracy and other performance measures."

ML-3: Diabetes Prediction using KNN

What is KNN?

KNN = K-Nearest Neighbors.

It checks the nearest data points and predicts the class using majority voting.

Example:

K = 5

- 3 people have diabetes.
- 2 people don't have diabetes.

Prediction = Diabetes (1).

. Most important viva questions

Question	Answer
KNN full form?	K-Nearest Neighbors
What is Outcome?	Target column

What is K?	Number of nearest neighbors
What is StandardScaler?	Scales data
What is Accuracy?	Correct predictions
What is Precision?	Out of predicted positives, how many are correct
What is Recall?	Out of actual positives, how many are found
What is Error Rate?	Percentage of wrong predictions
What is Euclidean Distance?	Straight-line distance between points
What is Supervised Learning?	Learning from labelled data

ML-4: Sales Data Clustering using K-Means

2. What is Clustering?

Clustering means making groups of similar data.

Example:

We have 100 customers.

- Cluster 1 = Low spending customers.
- Cluster 2 = Medium spending customers.
- Cluster 3 = High spending customers.

3. What is K-Means?

K-Means is an Unsupervised Machine Learning algorithm.

It divides data into K different clusters.

- K = Number of clusters.
- Centroid = Center point of a cluster.

. What is Elbow Method?

Elbow Method helps us find the best number of clusters.

It uses WCSS (Within-Cluster Sum of Squares).

Simple meaning:

It measures how close data points are to their cluster center.

We calculate WCSS for different K values.

Then we draw a graph.

Where the graph looks like an elbow, that point gives us the optimal K.

Examiner asks: What is WCSS?

Ans: It measures the sum of squared distances between data points and their cluster centroids.

7. Most important viva questions

Question	Baby English answer
What is K-Means?	Algorithm used to make clusters.
What is Clustering?	Grouping similar data.
What is Unsupervised Learning?	Learning without labelled output.
What is K?	Number of clusters.
What is Centroid?	Center point of a cluster.
What is Elbow Method?	Method to find optimal K.
What is WCSS?	Sum of squared distances from points to their cluster centers.
What is StandardScaler?	Brings features to a similar scale.
What is Inertia?	WCSS value calculated by K-Means.
What is `fit_predict()`?	Fits the model and assigns cluster labels.

We load sales data, scale numerical features, use the Elbow Method to find the optimal number of clusters, and then apply K-Means to group similar sales records."

ML-5: Gradient Descent Algorithm

Important: Formula + Iterations + Learning Rate

1. Aim

To find the local minimum of a function using the Gradient Descent algorithm.

Given function:

$$y = (x + 3)^2$$

Starting value:

$$x = 2$$

Examiner ko bolo:

"Sir, in this practical, we use Gradient Descent to find the minimum value of a function by updating x again and again."

2. What is Gradient Descent?

Gradient Descent is an optimization algorithm.

It helps us find the minimum value of a function.

Simple example:

Imagine a ball on a hill.

The ball moves downwards and reaches the lowest point.

Similarly, Gradient Descent moves towards the minimum value.

3. Understand our function

Given:

$$y = (x + 3)^2$$

Starting value:

$$x = 2$$

The minimum value occurs at:

$$x = -3$$

$$y = (-3 + 3)^2 = 0$$

So our answer should move towards x = -3.

First understand the practical

MetaMask is a crypto wallet. It helps us store and manage Ethereum accounts and interact with blockchain applications.

Important viva questions

Question	Baby English answer
1. What is MetaMask?	MetaMask is a crypto wallet used to manage Ethereum accounts.
2. What is Cryptocurrency?	Digital money that works using cryptography.
3. What is Ethereum?	Ethereum is a blockchain platform.
4. What is Ether?	Ether (ETH) is the cryptocurrency of Ethereum.
5. What is Gas Fee?	Fee paid to process a transaction on Ethereum.
6. What is Gas?	It is the unit used to measure computational work on Ethereum.
7. What is a Transaction?	Transfer of data or crypto on blockchain.
8. What is Blockchain?	A digital record book that stores transactions.
9. What is a Browser Extension?	A small software added to a browser.
10. Why do we use MetaMask?	To manage crypto and connect with blockchain applications.

Most important: Gas Fee

Gas fee is the fee paid for performing transactions on Ethereum.

Formula:

\[Gas\ Fee=Gas\ Used\times Gas\ Price \]

Example:

- Gas used = 21,000
- Gas price = 10 Gwei

Gas fee = 0.00021 ETH.

Examiner: Why do we pay gas fees?

Ans: To pay for the computational work done by the Ethereum network.

BC-2: Create Your Own Wallet Using MetaMask

First understand the practical

We create our own crypto wallet using MetaMask.

A wallet helps us receive, send and manage cryptocurrency.

Important viva questions

Question	Baby English answer
1. What is a Crypto Wallet?	It is used to manage cryptocurrency accounts.
2. What is a Public Key?	It is used to identify an account and receive transactions.
3. What is a Private Key?	It is a secret key used to authorize transactions.
4. What is Wallet Address?	It is a public address used to send and receive crypto.
5. What is Secret Recovery Phrase?	A secret set of words used to recover a wallet.
6. How many words are in a standard MetaMask recovery phrase?	Usually 12 words.
7. Can we share our private key?	No, never share it.
8. What happens if we lose our recovery phrase?	We may permanently lose access if we cannot recover the wallet another way.
9. What is a Public Address?	It is like an account number.
10. What is the difference between public and private key?	Public key is shared; private key is kept secret.

Important difference

Public Address	Private Key
Used to receive crypto	Used to authorize transactions
Can be shared	Must be secret
Like account number	Like secret password/signature key

Examiner: What is a Seed Phrase?

Ans: It is a group of secret words used to recover a crypto wallet.

Remember: Never share your Secret Recovery Phrase with anyone.

BC-3: Smart Contract for Customer Bank Account

First understand the practical

We create a smart contract for a bank account.

It performs 3 operations:

1. Deposit money.
2. Withdraw money.
3. Show balance.

What is Smart Contract?

A smart contract is a program stored on the blockchain that automatically executes when its conditions are met.

Example:

If a customer deposits 100 ETH, the contract updates the account balance.

Important viva questions

Question	Baby English answer
1. What is Smart Contract?	A program stored on blockchain that executes automatically.
2. Which language is used to write Ethereum smart contracts?	Solidity.
3. What is Solidity?	Programming language used to write smart contracts.
4. What is Remix IDE?	Online tool used to write, compile and deploy smart contracts.
5. What is Ethereum Virtual Machine (EVM)?	It executes smart contracts on Ethereum.
6. What is Deploy?	Putting a smart contract on blockchain.
7. What is Compile?	Checking and converting code into bytecode.
8. What is uint?	An unsigned integer data type.
9. What is msg.sender?	Address of the person or account calling the contract.
10. What is require()?	It checks a condition before continuing.
11. What is payable?	It allows a function to receive Ether.

12. What is mapping?	It stores data in key-value pairs.
13. What is public?	It allows a variable or function to be accessed publicly.
14. What is view?	Function that reads data without changing blockchain state.
15. What is gas?	Computational cost of executing operations.

Three operations

Operation	Meaning
Deposit	Add money to account
Withdraw	Remove money from account
Show Balance	Display account balance

Baby English: It tells who is calling the contract.

Examiner's tricky questions

Q. What is a Test Network?

A test network is used to test smart contracts without using real money.

Q. What is Sepolia?

Sepolia is an Ethereum test network.

Q. What is the difference between Mainnet and Testnet?

- Mainnet = Real blockchain, real transactions.
- Testnet = Testing blockchain, test Ether.

Q. Can we change a deployed smart contract?

Generally, deployed contract code is immutable. Changes usually require deploying a new contract or using an upgradeable contract design.

BC-4: Survey Report on Types of Blockchains and Real-Time Use Cases

First understand the practical

We study different types of blockchains and their real-world applications.

There are mainly 4 types.

1. Public Blockchain

Anyone can join and participate.

Example: Bitcoin, Ethereum.

Use cases:

- Cryptocurrency.
- Money transfer.
- Decentralized applications.

2. Private Blockchain

Only selected people can access it.

Example: Private company blockchain.

Use cases:

- Company records.
- Internal banking.
- Supply chain management.

3. Consortium Blockchain

It is controlled by a group of organizations.

Use cases:

- Banking networks.
- Multiple hospitals sharing records.
- Supply chain between companies.

4. Hybrid Blockchain

It combines public and private blockchain features.

Use cases:

- Healthcare.
- Government records.
- Business applications.

Important viva questions

Question	Baby English answer
1. How many main types of blockchain?	Four main types.
2. What is Public Blockchain?	Anyone can participate.
3. What is Private Blockchain?	Only authorized people can access it.
4. What is Consortium Blockchain?	Controlled by multiple organizations.
5. What is Hybrid Blockchain?	Combination of public and private blockchain.
6. What is Bitcoin?	A digital cryptocurrency.
7. What is Decentralization?	Control is not held by one central authority.
8. What is Transparency?	Transactions can be viewed by permitted participants.
9. What is Immutability?	Recorded data is very difficult to change.
10. What is Consensus?	Method used by network participants to agree on transactions.
11. What is a Node?	A computer participating in a blockchain network.
12. What is a Block?	A group of recorded transactions.
13. What is a Hash?	A fixed-length digital value generated from data.
14. What is a Distributed Ledger?	A record shared across multiple computers.

Important difference

Public	Private
Open participation	Restricted participation
More decentralized	Controlled by an organization
Example: Bitcoin	Example: Company private blockchain

Examiner: What are real-life uses of blockchain?

Ans:

1. Banking – Secure transactions.

2. Healthcare – Medical records.
3. Supply Chain – Tracking products.
4. Voting – Recording votes.
5. Education – Verifying certificates.